

Homework Grading Report

Student:	Student_10
Assignment:	Midterm Exam (mt13)
Date:	February 08, 2026
Grade:	119.3 / 125 (95%) A

Section Breakdown

Section	Points	Earned	Notes
Part 1: R Basics and Data Import	10	10.0	All tasks completed
Part 2: Data Cleaning	15	7.5	Missing: drop_na(), IQR()
Part 3: Basic Data Transformation	15	15.0	Missing: %>%()
Part 4: Advanced Transformation	15	15.0	All tasks completed
Part 5: Data Reshaping	15	15.0	All tasks completed
Part 6: Combining Datasets	15	15.0	All tasks completed
Part 7: String Manipulation & Date/Time	20	20.0	All tasks completed
Part 8: Advanced Wrangling	15	15.0	All tasks completed
Part 9: Reflection Questions	5	5.0	All tasks completed
TOTAL	125	119.3	95%

Detailed Feedback

What You Did Well

- Your code demonstrates excellent understanding of R fundamentals and data manipulation techniques throughout all sections.
- You consistently use the pipe operator (%) effectively in transformation tasks, making your code readable and maintainable.
- Your data cleaning approach with na.omit() correctly removes missing values without introducing errors.
- Your statistical analysis in Parts 4 and 8 shows proper application of summarise() with group_by() for business intelligence calculations.
- Your data reshaping using pivot_wider() correctly transforms the data from long to wide format with appropriate column naming.
- Your reflection correctly identifies the business relevance of join types, linking left_join to retaining all customers for churn analysis and showing you understand how data integration supports

retention strategies.

What to Fix

- 1. WHAT: Your answer to the missing:** values question lacks specific evidence from the dataset. **WHY:** Without showing how many records were examined, the claim of "no missing values" cannot be verified, reducing confidence in data quality. **HOW:** Include a line of code (e.g., `sum(is.na(sales_data))`) and report the count of NAs you found; if none, state the exact result. **EXAMPLE:** The solution prints "Total NAs in Revenue: 0", confirming the dataset is clean.
- 2. WHAT:** In the regional and category summary response you did not cite any concrete metric or pattern. **WHY:** Business stakeholders need tangible figures (e.g., "North region generated \$1.2M, 15% above average") to act on the analysis. **HOW:** After grouping, extract a key statistic such as highest-selling region or category and mention the value in your reflection. **EXAMPLE:** The model solution notes "West region had the highest average order value of \$342", which you could reference.
- 3. WHAT: Your data:** reshaping answer repeats the same wide-format example, making the explanation redundant. **WHY:** Distinct scenarios demonstrate deeper understanding of when each format is optimal, which is essential for efficient workflow design. **HOW:** Keep the wide-format example for monthly sales reporting, and add a separate long-format scenario such as feeding transaction-level data into a time-series forecasting model. **EXAMPLE:** The reference solution uses long format to model weekly purchase frequency per customer.
- 4. WHAT:** The KPI discussion does not name a specific metric you created in Lesson 8. **WHY:** Naming a concrete KPI (e.g., "Customer Lifetime Value") shows you can translate wrangling steps into actionable business measures. **HOW:** In your reflection, state the exact KPI you calculated, the formula you used, and why it matters for strategic decisions. **EXAMPLE:** The exemplar answer describes creating " $\text{Profit_Margin} = (\text{Revenue} - \text{Cost}) / \text{Revenue}$ " and explains how it flags low-margin products.

Reflection & Critical Thinking

You noted that there were no missing values or outliers and explained why cleaning matters. The answer is accurate but lacks concrete examples from the dataset, such as how many rows would have been affected. Adding a brief description of the checks you performed would strengthen the response.

Your explanation of how grouped analysis reveals regional and category trends is correct and ties to marketing and inventory decisions. However, you did not cite any specific metric or pattern you observed in the data. Including a concrete insight, like which region had the highest average sales, would make the answer more compelling.

You described the purposes of wide and long formats and gave business scenarios for each, which shows understanding of data reshaping. The response repeats the same example twice, reducing clarity. Streamlining the answer and providing a distinct second scenario, such as using long format for time-series modeling, would improve it.

Your comparison of `left_join` and `inner_join` correctly identifies the row-retention differences and appropriate business contexts. This answer is concise and accurate, demonstrating solid grasp of join logic. No further detail is needed here.

You selected Lesson 8 as the most valuable because it enables KPI creation, linking it to strategic decisions. The rationale is sound but could be deepened by naming a specific KPI you built, like profit margin per product. Providing that example would illustrate the skill's impact.

Instructor Assessment

You have answered every reflection prompt, which shows you understand the core concepts. However, most responses remain generic and would benefit from concrete numbers or code evidence drawn from your actual datasets. Strengthen your reflections by quoting specific results (e.g., counts of missing values, top-performing regions) and by naming the exact KPIs you built. Doing so will turn solid technical work into compelling business insight.

Key Takeaway

Complete all reflection questions with detailed, data-driven examples to demonstrate analytical depth.

In future assignments, embed short code snippets or output summaries within your narrative to substantiate claims.

Practice articulating specific business metrics derived from your wrangling steps, linking each metric to a clear decision-making context.